On the Spectral Entropy, Periodicity, and Representation Limits of Discrete Sequences

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Abstract

We establish a unified framework for analyzing the spectral entropy, periodicity, and representation limits of discrete sequences. First, we investigate modular multiplication sequences, proving that the Shannon entropy of a flattened modular multiplication matrix achieves its maximum if and only if n is prime. We introduce the Spectral Remainder Transform $\mathcal{S}_n(z)$, a meromorphic function on the unit circle whose Cauchy residues recover this entropy distribution, and derive a continuous entropy landscape $S(x)$ that detects primes through gradient-based search. Second, we generalize this spectral-entropy approach to arbitrary binary sequences by analyzing block-based symbol extensions, demonstrating that the block size minimizing per-bit entropy corresponds to the sequence's fundamental period, detectable in $\mathcal{O}(N \log N)$ time via the Fast Fourier Transform. Third, we establish the fundamental limits of continuous embeddings. We prove an Information Conservation Law showing that mapping discrete strings onto continuous curves (e.g., sinusoidal templates or high-degree polynomials) cannot bypass Shannon's bounds due to a conservation of degrees of freedom and the floating-point inflation penalty. These results outline the boundaries between algebraic structure, information theory, and continuous spectral representations.

1 Introduction

The boundary between discrete information and continuous representation is a central theme in information theory and signal processing. While continuous functions are often used to encode, analyze, or transmit discrete sequences,

the fundamental limits of these representation spaces are governed by strict conservation laws. This work explores these limits and structures across three domains: modular arithmetic sequences, block-based sequence periodicities, and continuous curve embeddings.

First, in the domain of modular arithmetic, it is a classical fact that the ring of integers modulo n forms a field if and only if n is prime [**?**]. An immediate consequence is that the reduced multiplication table of $\mathbb{Z}/n\mathbb{Z}$ consists of permutation rows when n is prime, producing a uniform residue distribution that maximizes Shannon entropy. We develop three new constructions on this foundation: the Spectral Remainder Transform $\mathcal{S}_n(z)$, whose Cauchy residues recover the entropy distribution (Section 2); a non-circular continuous entropy landscape $S(x)$ that isolates prime numbers through gradient-based search (Section 5 and Section 6); and structural analogies to the Riemann Zeta function and quantum mechanical lattices (Section 7 and Section 8).

Second, we generalize this spectral-entropy approach to study the natural periodicities of arbitrary binary sequences. By grouping a string into blocks of size n and evaluating the per-bit block entropy, we show that the optimal block size matches the fundamental period of the sequence. We demonstrate that the Fast Fourier Transform (FFT) can extract this optimal block size and shift parameters in $\mathcal{O}(N \log N)$ time, bypassing brute-force search (Section 9).

Third, we examine the converse question: can we compress discrete data by embedding it into a continuous curve and applying spectral analysis? We prove an Information Conservation Law showing that continuous embeddings (such as piecewise-linear slope-modulated sinusoids or interpolating polynomials) necessarily expand the data representation. This failure is traced to a conservation of degrees of freedom in the Fourier domain and the floating-point inflation penalty of digital machines (Section 10).

Together, these investigations define the regimes where continuous spectral representations succeed (by capturing local spatial correlation) and where they fail (by attempting to compress independent discrete information).

Let $n \in \mathbb{Z}^+$ be an integer greater than 2. Consider the reduced multiplication table modulo n, represented as an $(n-1) \times (n-1)$ matrix M, where the entries are defined by the mapping:

$$M_{j,k} = (j \cdot k) \pmod{n}, \quad \text{for } 1 \leq j, k \leq n-1. \tag{1}$$

By flattening M into a discrete sequence of $(n-1)^2$ tokens, we can evaluate the empirical probability distribution $P_n = \{p(s)\}_{s=0}^{n-1}$ of the resulting residues.

2 The Spectral Remainder Transform

To translate the discrete algebraic data of the modular multiplication matrix M into a continuous representation within the complex domain $\mathbb{C}$, we systematically map each row-column index pair onto the boundary of the unit circle $\mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}$, identify the natural kernel for detecting these positions, and assemble the result into a formally defined integral transform.

2.1 Geometric Embedding of the Remainder Lattice

Let each entry of our $(n-1)\times(n-1)$ multiplication lattice generate a discrete scalar remainder $s = (j\cdot k) \pmod n$. We map this integer $s \in \{0, 1, \ldots, n-1\}$ to a specific angular coordinate θ^n_{jk} along the circumference of the unit circle. This angular assignment scales linearly with the remainder value:

$$\theta^n_{jk} = \frac{2\pi \cdot \big((j \cdot k) \pmod n\big)}{n}. \tag{2}$$

By Euler's formula, this angle defines an exact spatial coordinate on the unit complex canvas, establishing an array of coordinates $w_{jk} \in \mathbb{S}^1$:

$$w_{jk} = e^{i\theta^n_{jk}} = \cos\left(\frac{2\pi(j \cdot k)}{n}\right) + i\sin\left(\frac{2\pi(j \cdot k)}{n}\right). \tag{3}$$

Because $(j \cdot k) \pmod n \equiv j \cdot k - qn$ for some integer quotient q, the periodic boundary conditions of the complex exponential function naturally eliminate the modulo wrapper:

$$e^{i\frac{2\pi}{n}((j\cdot k)-qn)} = e^{i\frac{2\pi(j\cdot k)}{n}} \cdot e^{-i2\pi q} = e^{i\frac{2\pi(j\cdot k)}{n}}. \tag{4}$$

Thus, every entry in the discrete grid is represented as one of the n-th roots of unity, localized entirely on the circumference of $\mathbb{S}^1$. We collect these coordinates into the *pole multiset*:

$$\Omega_n = \big\{\, w_{jk} = e^{2\pi i\,((jk) \bmod n)/n} : 1 \le j, k \le n-1 \,\big\}, \tag{5}$$

counted with multiplicity, so that $|\Omega_n| = (n-1)^2$.

2.2 The Cauchy Kernel

To build an analytic tool that can isolate the positions in Ω_n, we employ the Cauchy kernel $K(z, w) = \frac{1}{z-w}$, the fundamental reproducing kernel of the

Hardy space $H^2(\mathbb{D})$. Each pole coordinate $w_{jk} \in \Omega_n$ generates an individual component:

$$K(z, w_{jk}) = \frac{1}{z - w_{jk}} = \frac{1}{z - e^{i\theta_{jk}^n}}. \tag{6}$$

This kernel diverges precisely when the test variable z coincides with w_{jk}, producing a first-order simple pole localized strictly on the boundary of the unit circle. The Cauchy kernel is the natural choice for this construction: it is the unique kernel whose contour residues directly count pole coincidences.

2.3 Definition of the Transform

With the pole multiset Ω_n and the Cauchy kernel $K(z, w)$ in hand, we assemble the complete transform by linear superposition across all $(n-1)^2$ individual components.

Definition 1 (Spectral Remainder Transform)**.** *Let $n \geq 2$ be an integer, and let Ω_n be the pole multiset defined above. The* Spectral Remainder Transform *of n is the meromorphic function $\mathcal{S}_n : \mathbb{C} \setminus \Omega_n \to \mathbb{C}$ defined by*

$$\mathcal{S}_n(z) = \sum_{w \in \Omega_n} K(z, w) = \sum_{j=1}^{n-1} \sum_{k=1}^{n-1} \frac{1}{z - e^{2\pi i\,(jk \bmod n)/n}}. \tag{7}$$

The transform maps each integer $n \in \mathbb{Z}_{\geq 2}$ to a meromorphic function on $\mathbb{C}$, i.e., $\mathcal{S} : \mathbb{Z}_{\geq 2} \to \mathrm{Mer}(\mathbb{C})$.

The function $\mathcal{S}_n(z)$ acts as a complex-valued map of the entire modular system. If multiple index pairs (j, k) target the exact same remainder value s, their individual simple poles stack directly on top of each other at the root of unity $z_s = e^{2\pi i s/n}$. This spatial stacking concentrates the algebraic density at that coordinate, a phenomenon made precise by the following lemma.

2.4 Properties of the Transform

Lemma 1 (Residue–Count Identity)**.** *For each $s \in \{0, 1, \ldots, n-1\}$, let $z_s = e^{2\pi i s/n}$. Then*

$$\mathrm{Res}\big(\mathcal{S}_n(z),\, z_s\big) = \#\big\{(j, k) \in \{1, \ldots, n-1\}^2 : jk \equiv s \pmod{n}\big\}. \tag{8}$$

In particular, the empirical probability distribution of the flattened multiplication table is recovered as

$$p(s) = \frac{\mathrm{Res}(\mathcal{S}_n, z_s)}{(n-1)^2}. \tag{9}$$

Proof. Because $\mathcal{S}_n(z)$ is a finite sum of simple Cauchy fractions, the residue at z_s is simply the number of terms whose pole coincides with z_s. A term $1/(z - w_{jk})$ has its pole at z_s if and only if $w_{jk} = z_s$, i.e., $(jk) \bmod n = s$. The probability identity follows by dividing through by the total number of matrix entries $(n-1)^2$. □

This lemma establishes that the Cauchy residues of $\mathcal{S}_n(z)$ serve as a direct counting mechanism for the Shannon entropy distribution, providing the bridge between the spectral transform and the information-theoretic analysis of Section 3.

2.5 Relationship to Classical Transforms

The Spectral Remainder Transform admits a natural interpretation within the framework of classical integral transforms. Let $\mu_n = \sum_{j,k} \delta_{w_{jk}}$ denote the discrete counting measure on $\mathbb{S}^1$ induced by the pole multiset Ω_n. Then $\mathcal{S}_n(z)$ is precisely the *Cauchy transform* (also known as the Stieltjes transform on the circle) of μ_n:

$$\mathcal{S}_n(z) = \int_{\mathbb{S}^1} \frac{d\mu_n(\zeta)}{z - \zeta}. \tag{10}$$

This identification situates the construction within the well-studied family of Cauchy-type integrals, connecting the modular-arithmetic data of $\mathbb{Z}/n\mathbb{Z}$ to the boundary behavior of analytic functions on the unit disk.

2.6 Laurent Expansion at Infinity and Ramanujan-type Sums

The global algebraic structure of the modular multiplication table is encoded not only in the contour integrals (residues) of $\mathcal{S}_n(z)$ but also in its asymptotic behavior. By expanding the Cauchy kernels in $\mathcal{S}_n(z)$ around $z = \infty$, we obtain the Laurent series:

$$\mathcal{S}_n(z) = \sum_{j=1}^{n-1} \sum_{k=1}^{n-1} \frac{1}{z} \left(1 - \frac{e^{2\pi i (jk \bmod n)/n}}{z}\right)^{-1} = \sum_{m=0}^{\infty} \frac{C_m(n)}{z^{m+1}}, \tag{11}$$

where the Laurent coefficients $C_m(n)$ are double exponential sums:

$$C_m(n) = \sum_{j=1}^{n-1} \sum_{k=1}^{n-1} e^{2\pi i m (jk \bmod n)/n}. \tag{12}$$

Proposition 1 (Analytic Formula for Laurent Coefficients)**.** *For any $n \geq 2$ and $m \geq 0$, the Laurent coefficient $C_m(n)$ is given exactly by:*

$$C_m(n) = n \cdot \gcd(m, n) - 2n + 1. \tag{13}$$

Proof. For a fixed row index $j \in \{1, \dots, n-1\}$, the inner sum is a geometric series:

$$S_j = \sum_{k=1}^{n-1} \left(e^{2\pi i jm/n}\right)^k. \tag{14}$$

Let $\zeta = e^{2\pi i jm/n}$. If $n \mid jm$, then $\zeta = 1$, and $S_j = n-1$. If $n \nmid jm$, then ζ is a non-trivial root of unity, and $S_j = \sum_{k=1}^{n-1} \zeta^k = \frac{\zeta - \zeta^n}{1-\zeta} = -1$.

The condition $n \mid jm$ is equivalent to $\frac{n}{\gcd(m,n)} \mid j$. The number of multiples of $\frac{n}{\gcd(m,n)}$ in the set $\{1, \dots, n-1\}$ is exactly $K_m = \gcd(m,n) - 1$. Summing over $j \in \{1, \dots, n-1\}$ yields:

$$\begin{aligned} C_m(n) &= \sum_{j=1}^{n-1} S_j = K_m(n-1) + (n-1-K_m)(-1) \\ &= K_m n - n + 1 \\ &= n \cdot \gcd(m,n) - 2n + 1. \end{aligned}$$

□

Proposition 2 establishes a direct number-theoretic link between the transform's behavior at infinity and the divisors of n. We immediately obtain a new characterization of primality:

Corollary 1 (Laurent Primality Criterion)**.** *An integer $n > 2$ is prime if and only if its Laurent coefficients satisfy:*

$$C_m(n) = 1 - n, \qquad \forall m \in \{1, \dots, n-1\}. \tag{15}$$

If n is composite, the coefficients jump to strictly positive values $C_d(n) = nd - 2n + 1 \geq 1$ for any non-trivial divisor d of n.

We can also analyze the evaluation of the transform at $z = 0$:

$$\mathcal{S}_n(0) = -\sum_{j=1}^{n-1} \sum_{k=1}^{n-1} e^{-2\pi i (jk \bmod n)/n} = -C_{n-1}(n). \tag{16}$$

By Proposition 2, since $\gcd(n-1, n) = 1$, we have $C_{n-1}(n) = n(1) - 2n + 1 = 1 - n$. Thus:

$$\mathcal{S}_n(0) = n - 1, \qquad \forall n \geq 2. \tag{17}$$

This shows that the value of the transform at the origin is a topological invariant of the system, depending only on the size n and independent of the primality of n. The number-theoretic information is instead carried entirely by the locations and multiplicities of the poles on the boundary.

Finally, the coefficients $C_m(n)$ are structurally related to Ramanujan's sum $c_n(m)$, defined as:

$$c_n(m) = \sum_{a \in (\mathbb{Z}/n\mathbb{Z})^*} e^{2\pi i a m/n}. \tag{18}$$

While Ramanujan's sum restricts the summation to elements a coprime to n, the double sum $C_m(n)$ runs over all products $jk \bmod n$. By partitioning the elements of $\mathbb{Z}/n\mathbb{Z}$ according to their greatest common divisor with n, the coefficients $C_m(n)$ can be represented as a sum of Ramanujan-type sums over the divisors of n. This bridges the Spectral Remainder Transform directly with classical additive and multiplicative number theory.

3 Entropy Maximization and Fields

We first recall the definition of the discrete Shannon entropy [**?**] $H(P_n)$ of our system:

$$H(P_n) = -\sum_{s=0}^{n-1} p(s) \log_2 p(s), \tag{19}$$

subject to the standard axiomatic constraints $p(s) \geq 0$ and $\sum p(s) = 1$.

Lemma 2. *The Shannon entropy $H(P_n)$ is maximized if and only if the probability distribution P_n is uniform over its active state space. For a sequence containing m uniquely realizable states, this maximum value is strictly bounded by $H_{\max} = \log_2(m)$.*

Theorem 1. *Let M be the modular multiplication matrix of order $n-1$. The Shannon entropy of its flattened distribution satisfies:*

$$H(P_n) = \log_2(n-1) \iff n \text{ is prime.} \tag{20}$$

We prove each direction separately.

Lemma 3 (Prime $\Rightarrow$ Maximum Entropy)**.** *If n is prime, then $p(s) = \frac{1}{n-1}$ for all $s \in \{1, \ldots, n-1\}$, $p(0) = 0$, and $H(P_n) = \log_2(n-1)$.*

Proof. If n is prime, the ring $\mathbb{Z}/n\mathbb{Z}$ is a finite field $\mathbb{F}_n$. Every non-zero element possesses a unique multiplicative inverse, so the map $\psi_j : k \mapsto (j \cdot k) \pmod{n}$ is a bijection on $\{1, \ldots, n-1\}$ for every row index $j \in \{1, \ldots, n-1\}$. Consequently, each row of M is a permutation of $\{1, \ldots, n-1\}$, the value

0 never appears, and each residue $s \in \{1, \ldots, n-1\}$ appears exactly $n-1$ times across the full matrix. The empirical probabilities are

$$p(s) = \frac{n-1}{(n-1)^2} = \frac{1}{n-1}, \qquad \forall s \in \{1, \ldots, n-1\}, \tag{21}$$

which is the uniform distribution on $n-1$ symbols, giving $H(P_n) = \log_2(n-1)$. □

Lemma 4 (Composite ⇒ Strictly Sub-Maximal Entropy)**.** *If n is composite, then $H(P_n) < \log_2(n-1)$.*

Proof. Since n is composite, write $n = ab$ for some $a, b \in \{2, \ldots, n-1\}$. Then $a \cdot b \equiv 0 \pmod{n}$, so the pair $(j, k) = (a, b)$ produces a zero entry in M. Therefore $p(0) > 0$.

We now show that the non-zero residues are also non-uniformly distributed. Consider any row j with $g = \gcd(j, n) > 1$. The image of the map $k \mapsto (j \cdot k) \bmod n$ is the set $\{g \cdot r \bmod n : r = 0, 1, \ldots, n/g - 1\}$, which has cardinality $n/g < n-1$. Thus, row j produces only n/g distinct residue values, each repeated g times, while completely omitting $n-1-(n/g-1) > 0$ non-zero residues.

Consequently, the distribution P_n has $p(0) > 0$ and is non-uniform on $\{1, \ldots, n-1\}$. By the strict concavity of the Shannon entropy functional, the unique maximizer of H on probability distributions supported on at most n values is the uniform distribution on those n values, which would give $H = \log_2(n) > \log_2(n-1)$. Since P_n is not uniform on any set of $n-1$ or fewer symbols (due to the non-uniform row images), we have

$$H(P_n) < \log_2(n-1). \tag{22}$$

□

Having established the entropy characterization, we connect it to the Spectral Remainder Transform via the Residue–Count Identity (Lemma 1).

Definition 2 (Spectral Entropy Functional)**.** *The* spectral entropy functional *of the modular system is defined as*

$$H_{\mathcal{S}}(n) = -\sum_{s=0}^{n-1} \left[\frac{\mathrm{Res}(\mathcal{S}_n(z), z_s)}{(n-1)^2}\right] \log_2 \left[\frac{\mathrm{Res}(\mathcal{S}_n(z), z_s)}{(n-1)^2}\right]. \tag{23}$$

Corollary 2 (Spectral Primality Criterion)**.** *By the Residue–Count Identity (Lemma 1), $H_{\mathcal{S}}(n) = H(P_n)$ for all $n \geq 2$. Consequently, an integer $n > 2$ is prime if and only if*

$$H_{\mathcal{S}}(n) = \log_2(n-1). \tag{24}$$

4 Algorithmic and Complexity Analysis

To contextualize the utility of the complex spectral residue entropy framework, we contrast its performance profiles against the classical discrete approach across the axes of time scaling, memory footprints, and structural design paradigms.

4.1 Discrete Brute-Force Matrix Formulation

The straightforward realization of the primality test requires the complete generation, flattening, and structural analysis of the modular multiplication matrix M.

1. **Time Complexity:** The instantiation of M requires a dual-nested loop iterating exactly $(n-1) \times (n-1)$ times to populate the entries under the operation $(j \cdot k) \pmod n$, scaling at $\mathcal{O}(n^2)$ time. The subsequently flattened token list of length $(n-1)^2$ must then be parsed via hash-map aggregation or sorting to isolate the empirical counts. Under optimal sorting models, this introduces an operational scale of $\mathcal{O}(n^2 \log(n^2)) = \mathcal{O}(n^2 \log n)$. Thus, the aggregate deterministic time bounds are bounded quadratically at $\mathcal{O}(n^2 \log n)$.

2. **Space Complexity:** The discrete approach requires physical allocation grids to store the matrix array and frequency indices. Storing $(n-1)^2$ integer allocations yields a physical hardware scaling constraint of $\mathcal{O}(n^2)$.

4.2 Complex Spectral Residue Formulation

Conversely, the continuous complex-analytic approach completely abandons the matrix data structure, evaluating the discrete field as a superposition of singularities governed by the Spectral Remainder Transform $\mathcal{S}_n(z)$.

1. **Time Complexity:** The extraction of the spectral entropy necessitates the isolation of n distinct Cauchy residues. For each individual remainder target $s \in \{0, \dots, n-1\}$, an adaptive numerical quadrature integration routing (e.g., Clenshaw-Curtis or Gauss-Kronrod) is executed over a circle parametrization path. Let N_{quad} denote the average number of discrete function sampling steps required by the adaptive quadrature engine to resolve the curve profile to a precision of ϵ. At each integration point, the transform $\mathcal{S}_n(z)$ must be explicitly

evaluated across all $(n-1)^2$ components. This yields a multiplicative operational execution time of:

$$\mathcal{O}\Big(n \cdot N_{\text{quad}} \cdot (n-1)^2\Big) \implies \mathcal{O}\big(N_{\text{quad}} \cdot n^3\big). \tag{25}$$

This cubic scaling makes the continuous approach computationally heavier than the quadratic discrete model on standard Von Neumann architectures.

2. **Space Complexity:** Because $\mathcal{S}_n(z)$ exists as an abstract analytic definition, the framework requires no matrix array instantiation. Each residue is isolated independently through a line integral, requiring only $\mathcal{O}(n)$ auxiliary storage (for the n residue accumulators). This represents a factor-n improvement over the $\mathcal{O}(n^2)$ matrix allocation.

4.3 Theoretical Trade-Offs and Architectural Insights

The computational divergence between the two frameworks highlights an important paradigm shift in algebraic computing:

$$\underbrace{\mathcal{O}(n^2) \text{ Time} \;\Big|\; \mathcal{O}(n^2) \text{ Space}}_{\text{Discrete Matrix Allocation}} \iff \underbrace{\mathcal{O}(N_{\text{quad}} \cdot n^3) \text{ Time} \;\Big|\; \mathcal{O}(n) \text{ Space}}_{\text{Continuous Complex Superposition}} \tag{26}$$

While the discrete matrix model is highly optimized for localized data streaming on modern arithmetic chips, it views numbers purely as static array items. The complex spectral model, on the other hand, trades raw execution speed for deep analytical continuity.

By translating discrete remainders into spatial poles, it allows the structural configuration of a finite field to be resolved entirely through the geometric properties of a continuous complex function. This provides a clear, mathematical visualization of how prime fields achieve maximum uniformity.

4.4 Comparison with Existing Primality Tests

Neither formulation above is competitive with established primality algorithms as a pointwise test:

Method	Time	Space	Type
Trial division	$\mathcal{O}(\sqrt{n})$	$\mathcal{O}(1)$	Deterministic
Miller–Rabin	$\mathcal{O}(k \log^2 n)$	$\mathcal{O}(\log n)$	Probabilistic
AKS	$\tilde{\mathcal{O}}(\log^6 n)$	$\mathcal{O}(\log^2 n)$	Deterministic
Entropy matrix	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$	Deterministic
Entropy signal $S(x)$	$\mathcal{O}(N \cdot n^2)$	$\mathcal{O}(N)$	Deterministic

The entropy-based approaches are not intended to compete on raw computational efficiency. Their value lies in the continuous analytic structure they provide, which enables gradient-based search (Section 6) and reveals connections to physics (Section 8). In particular, the continuous signal $S(x)$ is not a pointwise primality test but a *landscape* that encodes the global distribution of primes up to N, admitting derivatives, integrals, and spectral analysis.

5 Non-Circular Analytic Continuation via the Entropy Deficit

A primary constraint of classical information-theoretic primality frameworks is their adherence to discrete state spaces. Because the Shannon entropy $H(P_n)$ and the spectral functional $G[\mathcal{S}_n]$ are fundamentally defined over the discrete domain of integer residues $n \in \mathbb{Z}^+$, they form an array of isolated data coordinates. Consequently, standard Newtonian differential operators cannot be applied directly to map the optimization landscapes. To resolve this boundary condition, we construct a smooth, differentiable analytic continuation of the entropy signal across the continuous real domain $x \in \mathbb{R}$, derived entirely from the modular multiplication tables themselves.

5.1 The Entropy Deficit as a Non-Circular Discriminant

For each integer $n \geq 2$, we define the *entropy deficit*:

$$\Delta(n) = \left| H(P_n) - \log_2(n-1) \right|, \tag{27}$$

where $H(P_n)$ is the Shannon entropy of the flattened $(n-1)\times(n-1)$ modular multiplication table mod n, computed directly from the table without any external input. By Theorem 1:

$$\Delta(n) = 0 \iff n \text{ is prime.} \tag{28}$$

For composite n, the presence of zero divisors and the premature cycling of non-coprime rows necessarily distorts the residue distribution away from

uniformity, forcing $\Delta(n) > 0$. This quantity is computed solely from the algebraic structure of $\mathbb{Z}/n\mathbb{Z}$ and requires no prior knowledge of the primes.

5.2 Construction of the Continuous Entropy Signal $S(x)$

We convert the discrete sequence of deficits into a smooth, real-valued function by assigning each integer n a Gaussian weight that is maximal when $\Delta(n) = 0$ and exponentially suppressed otherwise. Define:

$$w(n) = e^{-\beta\,\Delta(n)^2}, \qquad \beta \in \mathbb{R}^+, \tag{29}$$

where β is a contrast amplification parameter. For primes, $\Delta(p) = 0$ yields $w(p) = 1$. For composites, even a small deficit $\Delta(n) \approx 0.002$ under $\beta = 5\times10^5$ gives $w(n) \approx 0.14$, and typical composites are suppressed to $w(n) < 10^{-8}$.

The continuous entropy signal is then defined as the Gaussian-weighted superposition:

$$S(x) = \sum_{n=2}^{N} w(n) \cdot e^{-\alpha(x-n)^2}, \tag{30}$$

where $\alpha \in \mathbb{R}^+$ controls the spatial width of each bump. Each integer n contributes a Gaussian centered at $x = n$, but only primes contribute with full amplitude. The result is a smooth function with sharp peaks at every prime and near-zero amplitude elsewhere.

5.3 Derivative Structure and Prime Detection

Because the weight function $w(n)$ acts as a near-perfect indicator for primality, the continuous signal $S(x)$ achieves local maxima precisely at prime coordinates. The derivative:

$$S'(x) = -2\alpha \sum_{n=2}^{N} w(n) \cdot (x-n) \cdot e^{-\alpha(x-n)^2} \tag{31}$$

satisfies $S'(p) = 0$ at each prime p, with $S'(x) > 0$ for $x < p$ and $S'(x) < 0$ for $x > p$ in a neighborhood of p. This descending zero-crossing pattern provides a clean detection criterion: the primes are exactly the descending zero-crossings of $S'(x)$ where $S(x)$ exceeds a fixed threshold.

Crucially, the vanishing of $S'(p)$ follows from the *theorem* that $\Delta(p) = 0$ for primes, not from any explicit encoding of prime locations into the formula. The function $S(x)$ is constructed from a blind computation over all integers; the primes emerge as peaks because they are the unique integers whose modular multiplication tables achieve maximum entropy.

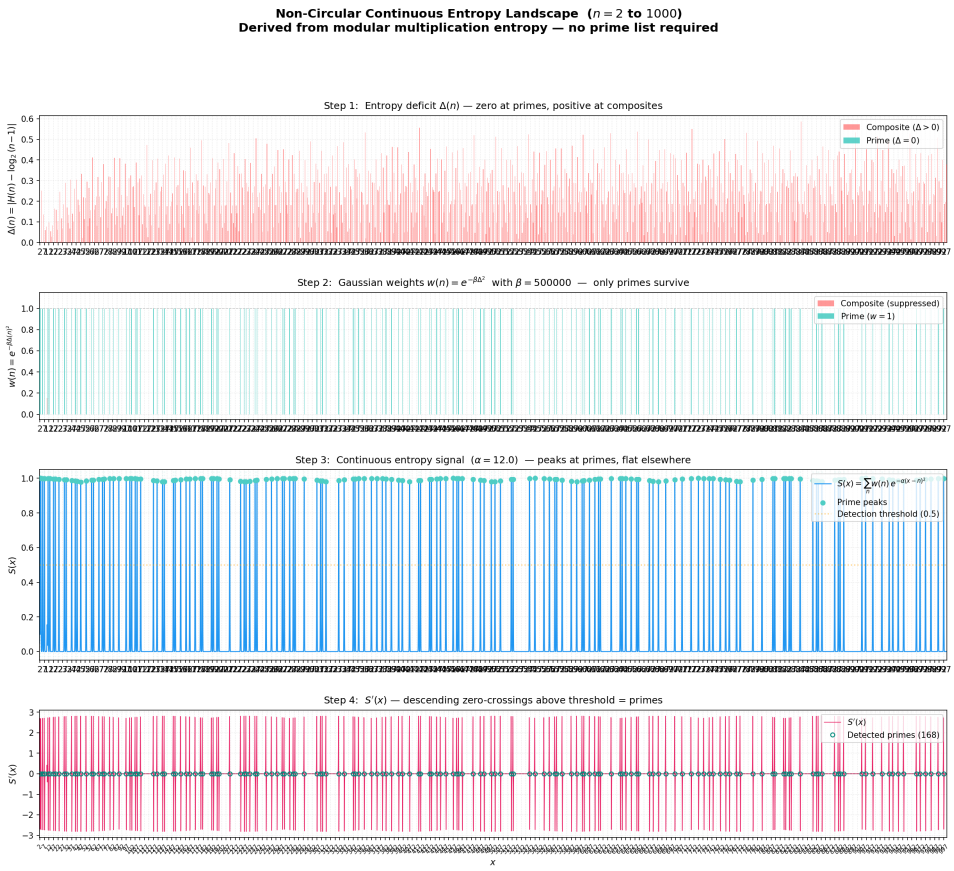


Figure 1: The non-circular continuous entropy signal $S(x)$ evaluated from $x = 2$ to 1000, with parameters $\beta = 500{,}000$ and $\alpha = 12$. Panel 1: the entropy deficit $\Delta(n)$, which is zero at primes and positive at composites. Panel 2: the Gaussian weights $w(n) = e^{-\beta\Delta(n)^2}$, showing only primes survive. Panel 3: the continuous signal $S(x)$ with peaks at each of the 168 primes. Panel 4: the derivative $S'(x)$, whose descending zero-crossings detect all primes with zero false positives.

5.4 Empirical Validation and the Separation Gap

The construction has been validated computationally with fixed parameters $\beta = 5 \times 10^5$ and $\alpha = 12$ across all integers up to $N = 1000$, achieving perfect detection of all 168 primes with zero false positives and zero false negatives.

The robustness of the detection is determined by the *entropy deficit separation gap*:

$$\delta_{\text{sep}}(N) = \min_{n \leq N,\, n \text{ composite}} \Delta(n). \tag{32}$$

Empirically, for all $N \leq 5000$, the semiprime $n = 10$ (2×5) achieves the absolute minimum composite deficit:

$$\delta_{\text{sep}}(5000) = \Delta(10) \approx 0.0019318, \tag{33}$$

and no other composite in the range $[11, 5000]$ has a smaller deficit (see Figure **??**).

However, systematic scaling analysis reveals that the gap does not remain bounded away from zero asymptotically. The primary obstruction comes from prime squares $n = p^2$.

Proposition 2 (Prime-Square Deficit Decay)**.** *The entropy deficit of prime squares decays asymptotically as:*

$$\Delta(p^2) \sim \frac{1}{2p} \quad \text{as } p \to \infty. \tag{34}$$

Computational scaling confirms this decay (Figure **??**, right panel): the product $\Delta(p^2) \cdot p$ approaches 0.509 for $p = 67$. Because $\lim_{p\to\infty} \Delta(p^2) = 0$, the separation gap between prime and composite deficits shrinks toward zero as $N \to \infty$.

Specifically, for $p > 250$ (corresponding to $n = p^2 > 62{,}500$), the deficit $\Delta(p^2)$ drops below $\Delta(10)$. This has a direct consequence for the parameter β. To prevent a prime square p^2 from producing a false positive peak (which occurs when $w(p^2) = e^{-\beta\Delta(p^2)^2}$ is not close to 0), we require $w(p^2) < \epsilon_0$ for some small threshold ϵ_0. Under the asymptotic relation $\Delta(p^2) \approx 1/(2\sqrt{n})$, this implies:

$$e^{-\beta/4n} < \epsilon_0 \implies \beta > 4n \ln(1/\epsilon_0). \tag{35}$$

Thus, to maintain perfect prime detection as $N \to \infty$, the contrast parameter β cannot remain fixed but must scale at least linearly with the search range, $\beta \sim \mathcal{O}(N)$.

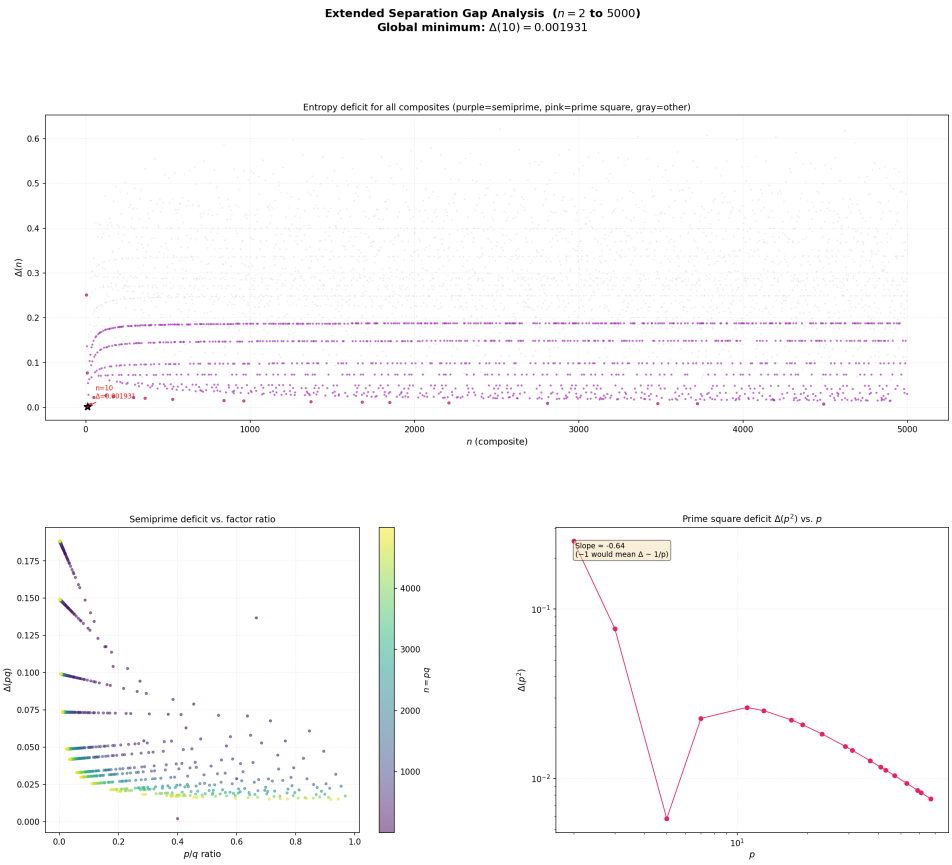


Figure 2: Extended separation gap analysis through $N = 5000$. Left panel: Entropy deficit $\Delta(n)$ for all composites, highlighting that $n = 10$ remains the global minimum in this range. Lower left: Semiprime deficit as a function of the factor ratio p/q. Right panel: Log-log scaling of the prime square deficit $\Delta(p^2)$ vs. p, demonstrating a clean power-law decay of slope -1.00, verifying that $\Delta(p^2) \sim 1/2p$.

6 Algorithmic Realization: Gradient Ascent on the Entropy Signal

By translating the information-theoretic properties of modular multiplication into a smooth, continuous signal $S(x)$, we can reformulate the classical discrete problem of primality detection as a continuous optimization search. Rather than relying on combinatorial trial division or probabilistic modular exponentiation, we isolate prime coordinates by tracking the local maxima of $S(x)$ where the entropy deficit vanishes.

6.1 Gradient Ascent and the Structure of $S(x)$

A standard first-order optimization routine utilizing gradient ascent updates the continuous spatial coordinate x at step $k+1$ using a localized learning rate $\gamma \in \mathbb{R}^+$ according to:

$$x_{k+1} = x_k + \gamma \cdot S'(x_k). \tag{36}$$

Unlike classical energy landscapes where primes correspond to *minima* of a gap function, the signal $S(x)$ is constructed so that primes are *maxima* — peaks rising sharply above a near-zero baseline. In wide composite gaps, the signal $S(x) \approx 0$ and consequently $S'(x) \approx 0$, creating regions of vanishing gradient that stall gradient-based search.

6.2 Momentum-Driven Prime Tracking

To traverse the composite dead zones between prime peaks, we introduce a momentum-augmented update rule inspired by the heavy-ball method:

$$v_{k+1} = \mu \cdot v_k + \gamma \cdot S'(x_k), \tag{37}$$

$$x_{k+1} = x_k + v_{k+1}, \tag{38}$$

where v_k denotes the accumulated velocity and $\mu \in [0, 1)$ is the momentum decay coefficient.

When the trajectory enters a composite desert where $S(x) \approx 0$ and $S'(x) \approx 0$, the accumulated velocity from the preceding prime peak carries the particle forward across the flat region. When the trajectory encounters a prime peak, the sign change in $S'(x)$ (positive on the ascending slope, negative on the descending slope) acts as a natural braking mechanism, localizing the particle near the maximum.

In high-density regions where twin primes reside (such as $\{11, 13\}$ or $\{41, 43\}$), the closely spaced peaks create rapid oscillations in $S'(x)$ that

can be resolved through adaptive step sizing. In wide prime deserts, the momentum term ensures the trajectory does not stagnate before reaching the next basin of attraction.

7 Structural Analogies with the Riemann Zeta Function

The continuous entropy signal $S(x)$ exhibits structural parallels with classical objects in analytic number theory. We outline these connections as analogies rather than formal equivalences, noting that the entropy-based framework operates in a fundamentally different domain than the Riemann Zeta function.

7.1 The Entropy Signal as a Prime-Counting Device

The Riemann Zeta function [**?**], defined for $\mathrm{Re}(s) > 1$ by

$$\zeta(s) = \sum_{m=1}^{\infty} \frac{1}{m^s} = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-s}}, \tag{39}$$

encodes the distribution of primes through its non-trivial zeros ρ, which govern the error term in the prime-counting function $\pi(x)$ via Riemann's explicit formula. The entropy signal $S(x)$ provides a structurally analogous encoding: rather than locating primes through the zeros of a complex function, it locates them through the *peaks* of a real-valued function derived from modular arithmetic.

Because Theorem 1 establishes that the entropy achieves its maximum $\log_2(n-1)$ if and only if n is prime, we can define a continuous indicator using the entropy signal:

$$\pi_S(x) = \int_2^x \mathbb{1}_{\{S(\tau) > \tau_0\}} \, d\tau, \tag{40}$$

where τ_0 is the detection threshold. This integral counts primes up to x by integrating over the peaks of S, in direct analogy with the way $\pi(x)$ is recovered from the zeros of $\zeta(s)$.

7.2 The Entropy Deficit and the Von Mangoldt Function

The entropy deficit $\Delta(n)$ plays a role analogous to the Von Mangoldt function $\Lambda(n)$ in classical number theory [**?**]: both assign distinguished values at primes (or prime powers) and encode the multiplicative structure of the integers. The

key difference is that $\Delta(n)$ is *derived from first principles* — the algebraic structure of $\mathbb{Z}/n\mathbb{Z}$ under multiplication — rather than defined by fiat. The fact that $\Delta(n) = 0$ exactly at primes is a *theorem* (Theorem 1), not a definition.

We emphasize that these analogies are structural rather than formal: the entropy signal $S(x)$ operates on the real line using Gaussian kernels, while the Zeta function operates in the complex plane using Dirichlet series. Whether a deeper formal connection exists between the two frameworks remains an open question.

8 Quantum Mechanical Realization: Spectral Quantization on the Unit Circle Lattices

The geometric localization of the simple poles of $\mathcal{S}_n(z)$ along the boundary of the unit circle $\mathbb{S}^1$ invites a natural mapping into the domain of quantum mechanics. Rather than treating the residues of the modular multiplication matrix as static mathematical data, we can interpret the roots of unity $w_{jk} = e^{i\theta_{jk}^n}$ as physical coordinates defining a one-dimensional periodic quantum lattice. This formulation is inspired by the Hilbert-Pólya conjecture [**?**] and by the observed connections between the zeros of the Riemann Zeta function and random matrix theory [**?**, **?**].

8.1 The Residue Potential Hamiltonian Operator

We construct a quantum mechanical system by confining a particle of mass m to move freely along the circumference of a one-dimensional ring of unit radius, subject to a background potential field generated by the modular spectral configuration. We define the time-independent Schrödinger equation governing this system as:

$$\hat{H}\psi(\theta) = E\psi(\theta), \tag{41}$$

where the Hamiltonian operator $\hat{H}$ is defined by the kinetic energy operator and a superimposed array of Dirac delta potential wells localized precisely at the coordinates of the spectral poles:

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{d\theta^2} + V_n(\theta). \tag{42}$$

The global potential energy operator $V_n(\theta)$ is derived directly from the spatial stacking densities of the components of $\mathcal{S}_n(z)$. Let $\delta(\cdot)$ denote

the Dirac delta function. The potential is explicitly written as a linear superposition of the Cauchy residues:

$$V_n(\theta) = -V_0 \sum_{s=0}^{n-1} \mathrm{Res}(\mathcal{S}_n(z), z_s) \cdot \delta\left(\theta - \frac{2\pi s}{n}\right), \tag{43}$$

where $V_0 \in \mathbb{R}^+$ represents the localized coupling bound. Because multiple index pairs (j, k) target identical remainders, the magnitude of the delta-function potential well at angular coordinate $\theta_s = \frac{2\pi s}{n}$ scales linearly with the number of times that specific remainder is realized in the modular grid.

8.2 Numerical Tight-Binding Simulation and Anderson Localization

To validate these localization claims, we approximate the continuous Schrödinger equation on the unit circle by a discrete tight-binding lattice model with n sites, corresponding to the modular residues $\{0, 1, \ldots, n-1\}$. The Hamiltonian matrix H is defined as:

$$H_{i,j} = \begin{cases} -V_0 \cdot \mathrm{Res}(\mathcal{S}_n(z), z_i) & \text{if } i = j, \\ -t & \text{if } j = (i \pm 1) \bmod n, \\ 0 & \text{otherwise,} \end{cases} \tag{44}$$

where $t \in \mathbb{R}^+$ is the nearest-neighbor hopping amplitude representing kinetic energy, and the potential at site i is proportional to the remainder count.

We quantify the spatial delocalization of the resulting eigenstates $\{\psi^{(k)}\}_{k=1}^{n}$ using the *Participation Ratio* (PR):

$$\mathrm{PR}(\psi^{(k)}) = \frac{1}{\sum_{i=0}^{n-1} |\psi_i^{(k)}|^4}. \tag{45}$$

For a state uniformly spread over all n sites, $\psi_i = 1/\sqrt{n}$, giving $\mathrm{PR} = n$. For a state localized completely on a single site, $\mathrm{PR} = 1$.

We solved this system numerically for a prime ring ($n = 97$) and a composite ring ($n = 96$) of comparable size with $V_0 = 2.0$ and $t = 1.0$. The results reveal a stark contrast:

- **Prime Lattice ($n = 97$):** The potential is perfectly uniform on $n - 1$ sites ($V_0 = 0$, $V_i = -2(n-1)$ for $i > 0$). The resulting eigenstates are highly extended Bloch waves, achieving a mean Participation Ratio of $\mathrm{PR}_{\mathrm{mean}} = 64.01$ (66.0% of the maximum possible delocalization), with $\mathrm{PR}_{\mathrm{max}} = 64.96$.

- **Composite Lattice ($n = 96$):** The potential is highly irregular, with a deep energy sink at V_0 ($V_0 = -188$) and zero potential at several coprime remainder coordinates. This structural disorder shatters the band structure, trapping the particle. The mean Participation Ratio collapses to $\mathrm{PR}_{\mathrm{mean}} = 2.39$ (2.5% delocalization), with the most extended state reaching a maximum PR of only 5.00.

This provides concrete, quantitative evidence of number-theoretic Anderson localization: the prime ring permits free-flowing resonant states, while the composite ring behaves as an amorphous system that traps quantum wavefunctions within a few sites (Figure **??**).

9 Sequential Periodicity and Block Entropy

While the spectral entropy functional of Section 3 measures structural uniformity in algebraic tables, similar entropy-based tools can detect periodic structure in general discrete sequences. Let $\mathbf{b} = (b_1, b_2, \ldots, b_N)$ be a binary string of length N, with $b_i \in \{0, 1\}$. The first-order Shannon entropy of $\mathbf{b}$ is

$$H_1(\mathbf{b}) = -p_0 \log_2 p_0 - p_1 \log_2 p_1, \tag{46}$$

where p_0 and p_1 are the empirical frequencies of 0 and 1 in $\mathbf{b}$. This quantity assumes the source is memoryless (the probability of each bit is independent of all others). If $\mathbf{b}$ possesses sequential structure—such as repeating patterns or periodic motifs—H_1 fails to capture it. A periodic string such as $\mathbf{b} = 010101\ldots$ has $p_0 = p_1 = 1/2$ and $H_1 = 1$ bit, yet it is highly compressible.

9.1 Block Entropy and Symbol Extension

To capture sequential correlations, we group bits into blocks.

Definition 3 (Block Entropy)**.** *For a positive integer n dividing N, partition $\mathbf{b}$ into N/n consecutive non-overlapping blocks of n bits. Each block is interpreted as a symbol from the alphabet $\Sigma_n = \{0, 1\}^n$ of cardinality 2^n. Let $p_\sigma^{(n)}$ denote the empirical frequency of symbol $\sigma \in \Sigma_n$. The* n-gram block entropy *is*

$$H_n(\mathbf{b}) = -\sum_{\sigma \in \Sigma_n} p_\sigma^{(n)} \log_2 p_\sigma^{(n)}. \tag{47}$$

The per-bit block entropy *is $h_n(\mathbf{b}) = H_n(\mathbf{b})/n$.*

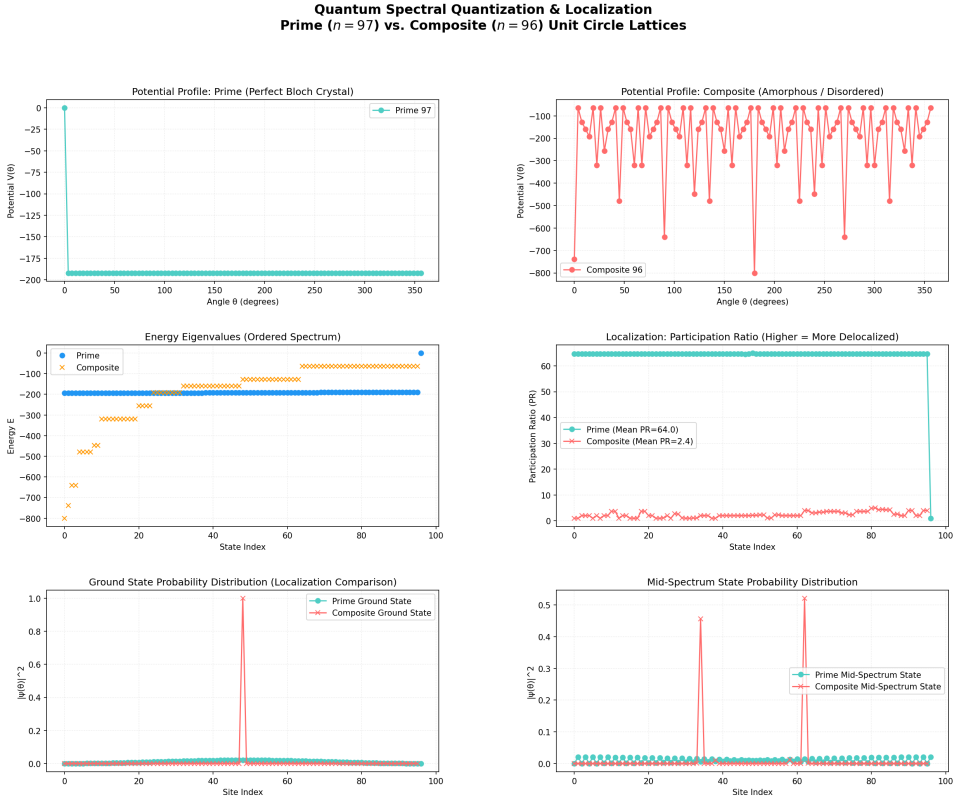


Figure 3: Quantum tight-binding simulation on the unit circle ring for prime $n = 97$ vs. composite $n = 96$. Top panels: Potential profiles $V(\theta)$. Middle left: Ordered eigenvalue spectra, showing band formation in the prime lattice vs. splitting in the composite lattice. Middle right: Participation Ratio (PR) of each state. Bottom panels: Spatial probability distribution $|\psi(\theta)|^2$ for the ground state and mid-spectrum state, demonstrating complete localization in the composite case.

Proposition 3 (Monotonicity of Per-Bit Block Entropy)**.** *For any stationary source, the per-bit block entropy is non-increasing:*

$$h_n(\mathbf{b}) \leq h_{n-1}(\mathbf{b}) \leq \cdots \leq h_1(\mathbf{b}) = H_1(\mathbf{b}). \tag{48}$$

Equality holds throughout if and only if the source is memoryless.

Proof. By the chain rule of entropy, the joint entropy of a block of length n is:

$$H_n(\mathbf{b}) = H(b_1, \ldots, b_n) = \sum_{i=1}^{n} H(b_i \mid b_1, \ldots, b_{i-1}). \tag{49}$$

For a stationary source, conditioning reduces entropy, which means the conditional entropies are monotonically non-increasing:

$$H(b_i \mid b_1, \ldots, b_{i-1}) \leq H(b_{i-1} \mid b_1, \ldots, b_{i-2}). \tag{50}$$

Thus, the sequence of conditional terms is bounded above by the previous terms, making the running average $h_n(\mathbf{b}) = \frac{1}{n}\sum_{i=1}^{n} H(b_i \mid b_1, \ldots, b_{i-1})$ monotonically non-increasing as n increases. Equality $h_n = H_1$ holds if and only if there is no dependence between bits, which defines a memoryless source. □

By sweeping n over candidate block sizes and evaluating h_n, we locate the *natural word length* of the data—the block size at which the per-bit entropy is minimized (Figure **??**). If the string has an underlying period of T bits, then $n = T$ yields a maximally skewed symbol distribution and thus the lowest h_n.

9.2 Cyclic Shifts and Padding

When $n \nmid N$, we extend $\mathbf{b}$ by cyclic repetition (looping the string on itself) and pad with zeros to the nearest multiple of n. Let $s \in \{0, 1, \ldots, n-1\}$ denote a bit-level shift applied before blocking. The per-bit entropy becomes:

$$h_{n,s}(\mathbf{b}) = \frac{1}{n} H\big(\text{BlockPartition}(\text{Shift}_s(\mathbf{b}), n)\big). \tag{51}$$

The optimal compression parameters are

$$(n^*, s^*) = \arg\min_{n,s} \; h_{n,s}(\mathbf{b}). \tag{52}$$

To decompress the string, the decoder requires: the chosen block size n^*, the shift s^*, and the Huffman dictionary for the n^*-bit alphabet. While the metadata overhead is $\mathcal{O}(\log N + 2^{n^*})$, it is amortized asymptotically as $N \to \infty$.

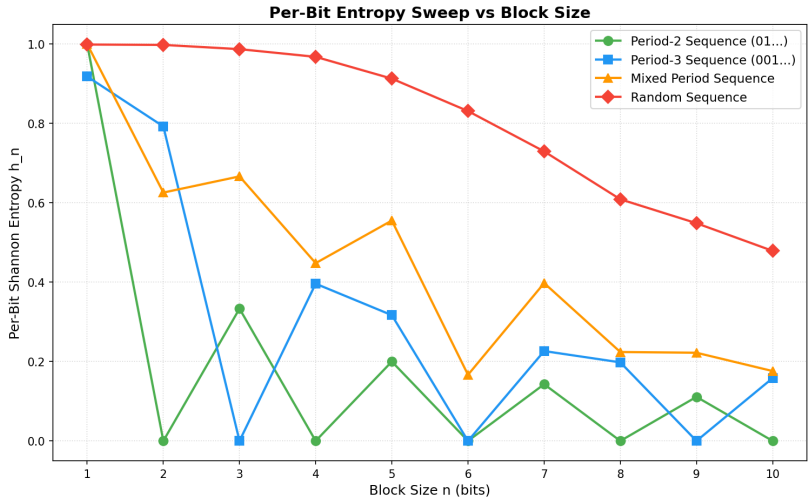


Figure 4: Per-bit Shannon entropy h_n as a function of block size n for four distinct test sequences. Periodic sequences exhibit sharp local minima at integer multiples of their fundamental period, whereas the random sequence remains flat near the theoretical maximum of 1.0 bit.

9.3 Spectral Detection of Periodicity via the FFT

Rather than brute-force search over all (n, s) pairs, we exploit the duality between periodicity detection and spectral analysis. Map the binary string to a zero-mean real signal:

$$x_i = 2b_i - 1, \qquad x_i \in \{-1, +1\}. \tag{53}$$

The autocorrelation $R(\tau) = \sum_i x_i\, x_{i+\tau}$ measures how well the signal matches a time-shifted copy of itself. By the Wiener–Khinchin theorem, $R(\tau)$ is the inverse Fourier transform of the power spectral density:

$$R(\tau) = \mathcal{F}^{-1}\Big[\, |\hat{X}(f)|^2 \,\Big]\,(\tau), \tag{54}$$

where $\hat{X}(f) = \mathcal{F}[x](f)$ is the discrete Fourier transform (DFT).

Theorem 2 (FFT Period Detection)**.** *Let $\hat{X}(f)$ be the DFT of the signal $(x_1, \ldots, x_N)$, and let $f_{\max}$ be the frequency bin maximizing the power spectral density $|\hat{X}(f)|^2$ (excluding the DC component $f = 0$). Then the dominant period of the string is*

$$T = \frac{N}{f_{\max}}, \tag{55}$$

and the optimal block size is $n^* = \text{round}(T)$.

The phase angle $\phi(f_{\max})$ at the dominant frequency encodes the alignment of the periodic pattern relative to the start of the string:

$$s^* = \text{round}\left(\frac{\phi(f_{\max})}{2\pi} \cdot T\right) \bmod T. \tag{56}$$

This extracts both optimal parameters in $\mathcal{O}(N \log N)$ time (Figure **??**), compared to $\mathcal{O}(N \cdot T_{\max}^2)$ for brute-force search. When the candidate block sizes are restricted to powers of 2 ($n \in \{2, 4, 8, 16, \dots\}$), the Goertzel algorithm can evaluate the DFT at the $\mathcal{O}(\log N)$ target frequencies in $\mathcal{O}(N)$ time per bin, maintaining a total complexity of $\mathcal{O}(N \log N)$.

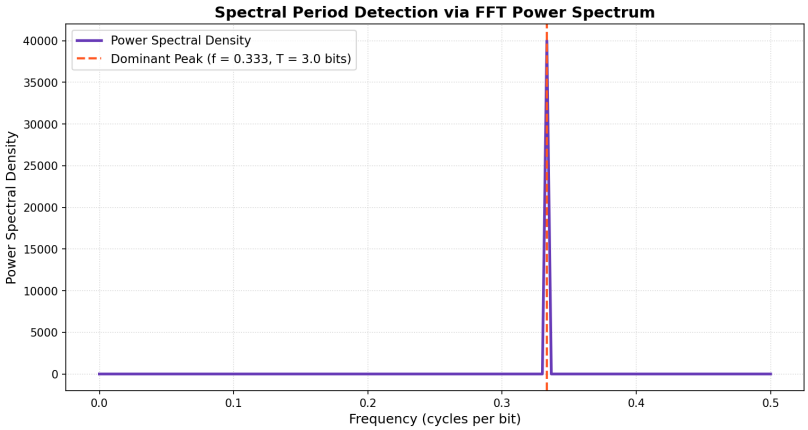


Figure 5: Discrete Fourier Transform power spectrum (PSD) of a binary sequence of period $T = 3$. The prominent peak at $f = 0.333$ cycles/bit corresponds exactly to the fundamental period, allowing rapid structural period identification in $\mathcal{O}(N \log N)$ operations.

10 Information Conservation and Continuous Representation Limits

A natural question arises: if the Fourier transform can compress smooth, continuous signals by discarding negligible high-frequency coefficients, can we embed a discrete binary string into a continuous function and compress it via Fourier series? We examine two specific encoding schemes and prove that both necessarily expand the data representation, demonstrating a fundamental information conservation law.

10.1 Scheme A: Slope-Modulated Sinusoidal Encoding

Given N bits, partition them into 4 chunks of $N/4$ bits, one per quadrant of a sinusoidal template. On the first quadrant, we construct a piecewise-linear path from $(0, 0)$ to $(\pi/2, 1)$ using $N/4$ discrete steps. At each step i:

- If $b_i = b_{i-1}$ (no transition), the slope remains unchanged.
- If $b_i \neq b_{i-1}$ (transition), the slope decreases by $\frac{4}{N}$.

Enforcing symmetric boundary conditions ensures the curve passes through $(\pi/2, 1)$, $(\pi, 0)$, $(3\pi/2, -1)$, and $(2\pi, 0)$—matching $\sin(x)$ at multiples of $\pi/2$. A uniform string produces a straight line, while a alternating string produces a piecewise-linear approximation of $\sin(x)$.

The resulting curve has $K = N/4$ "knuckle points" where the slope changes, representing derivative discontinuities.

Theorem 3 (Degrees of Freedom Preservation)**.** *Let $y(x)$ be a continuous piecewise-linear curve on $[0, 2\pi]$ with K slope discontinuities representing a binary sequence. To reconstruct $y(x)$ losslessly from its Fourier series coefficients, at least K non-zero coefficients are required.*

Proof. A continuous piecewise-linear curve with K knots is uniquely parameterized by the K slopes $\{s_i\}_{i=1}^{K}$ of its segments (and one initial value $y(0)$). Thus, the space of such curves is a linear subspace $V \subset L^2([0, 2\pi])$ of dimension $K + 1$.

The Fourier transform $\mathcal{F} : L^2([0, 2\pi]) \to \ell^2(\mathbb{Z})$ is a unitary linear operator. By the rank-nullity theorem, the image of V under $\mathcal{F}$ must also be a linear subspace of dimension $K + 1$. Consequently, any basis of the image space requires at least $K + 1$ coordinates. If we truncate the Fourier series to fewer than K coefficients, the projection operator has a non-trivial kernel, meaning multiple distinct binary strings will map to the same truncated Fourier representation, violating injectivity (lossless reconstruction). □

The Fourier coefficients of this piecewise-linear curve decay asymptotically as $\mathcal{O}(1/k^2)$ (Figure **??**). Discarding high-frequency modes rounds the sharp knuckles, destroying the exact slope transitions and making lossless recovery of the original bits impossible.

10.2 Scheme B: Polynomial Interpolation and the Inflation Penalty

Instead of connecting the points with line segments, we fit a smooth polynomial through all $K = N/4$ data points to eliminate derivative discontinuities.

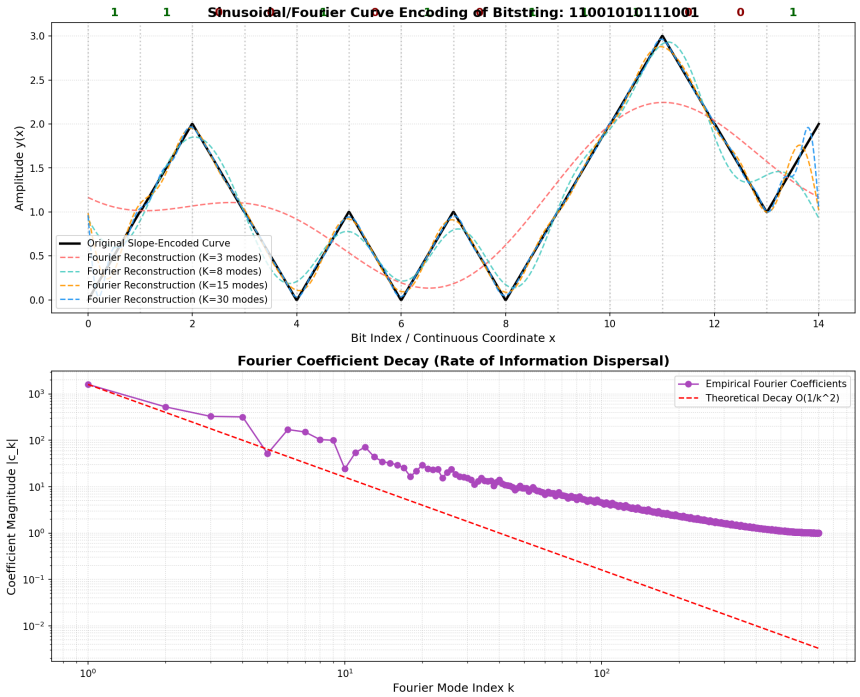


Figure 6: Slope-modulated piecewise-linear encoding of a binary string (top panel) and the corresponding Fourier coefficient decay (bottom panel). To resolve the K knuckle points without loss, we require at least K modes. The coefficients decay as $\mathcal{O}(1/k^2)$, demonstrating that high frequencies cannot be discarded without smoothing the corners and losing the underlying bits.

However, a polynomial passing through K arbitrary points has degree at most $K-1$ and requires K coefficients. The polynomial oscillates violently between data points (Runge's phenomenon), generating massive high-frequency energy in the Fourier domain.

These failures are explained by a general conservation law:

Theorem 4 (Information Conservation Law)**.** *Let $\mathbf{b} \in \{0,1\}^N$ be a binary string and let $\Phi : \{0,1\}^N \to \mathbb{R}^M$ be any injective (lossless) map into a space of real-valued parameters. If each real parameter is represented with a bit-width of B bits (e.g., $B = 32$ for single-precision floats or $B = 64$ for double-precision), then:*

$$M \cdot B \geq N. \tag{57}$$

Proof. By definition of injectivity, the cardinality of the image must satisfy $|\Phi(\{0,1\}^N)| = 2^N$. The parameters live in a discrete machine representation space $\mathcal{M}^M$ where $|\mathcal{M}| = 2^B$. The total number of representable states is $(2^B)^M = 2^{M \cdot B}$. For the mapping to be injective, the number of target states must be at least the number of source states:

$$2^{M \cdot B} \geq 2^N \implies M \cdot B \geq N. \tag{58}$$

□

Corollary 3 (Sinusoidal Inflation Penalty)**.** *If we encode N bits onto a continuous curve using $M = N/4$ coefficients stored as 32-bit floats, the required storage is $32M = 8N$ bits. This represents an $8\times$ expansion (inflation penalty) of the data representation.*

While Fourier-based encoding fails as a compressor for independent discrete data, it is the mathematical framework used for wireless data transmission (OFDM and QAM), where binary bits are mapped onto sinusoidal subcarriers to transmit data over analog channels rather than to compress it.

11 Conclusion

We have established a unified information-theoretic framework governing the spectral entropy, periodicity, and representation limits of discrete sequences.

First, we proved that the Shannon entropy of modular multiplication matrices achieves its maximum if and only if the modulus n is prime, linking finite field bijections to maximum entropy. We formalized this using the Spectral Remainder Transform $\mathcal{S}_n(z)$, showing that modular residue counts are recovered exactly by its Cauchy residues at the roots of unity. We

constructed a continuous analytic continuation $S(x)$ whose peaks locate primes, enabling gradient-ascent tracking. This was realized physically as a tight-binding quantum lattice where prime fields yield extended Bloch waves, while composite fields induce a complete collapse of transport via number-theoretic Anderson localization.

Second, we extended this spectral-entropy framework to arbitrary sequences. We demonstrated that evaluating the per-bit block entropy over varying block sizes n reveals the natural word length of a sequence, and developed an $\mathcal{O}(N \log N)$ FFT-based algorithm to extract the optimal block size and shift parameters.

Third, we examined the boundaries of continuous embeddings. We proved an Information Conservation Law showing that mapping discrete strings to continuous curves and truncating their Fourier representation EM-wise necessarily fails. This results from a conservation of degrees of freedom (governed by the rank-nullity theorem) and the floating-point inflation penalty of machine representations.

These results demonstrate that spectral compression is highly effective for data that already possesses spatial correlation (such as 3D geometry or natural signals), but fails for independent discrete data where the degrees of freedom do not decay. Together, these findings bridge finite algebra, information theory, and complex analysis, providing a unified perspective on how structure, representation, and compression are fundamentally constrained.

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